

GIT HANDS ON EXERCISES

HANDS ON 1

```
MINGW64:/c:/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git init
Initialized empty Git repository in C:/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject/.git/

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ echo "# Demo Project" > README.md

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ ls
README.md

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      README.md

nothing added to commit but untracked files present (use "git add" to track)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git add README.md
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git add .

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git status
On branch master
```

```
MINGW64:/c:/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git commit -m "Initial commit"
[master (root-commit) bc45ae7] Initial commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git log
commit bc45ae76a6c454d0b43654cb208c9113d2cc6b23 (HEAD -> master)
Author: kiru1711 <your-github-kiruthika.1171@gmail.com>
Date:   Tue Jul 28 11:49:31 2026 +0530

    Initial commit

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git config --global user.email "kiruthika.1171@gmail.com"

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git config --global --list
user.name=kiru1711
user.email=kiruthika.1171@gmail.com

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git remote -v

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git remote add origin https://github.com/kiru1711/DemoProject.git

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git remote -v
origin https://github.com/kiru1711/DemoProject.git (fetch)
origin https://github.com/kiru1711/DemoProject.git (push)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (master)
$ git branch -M main
```

```
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git branch
* main

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 237 bytes | 118.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kiru1711/DemoProject.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

OUTPUT:

The screenshot shows the GitHub interface for a repository named 'DemoProject' owned by 'kiru1711'. The repository is a public repository with 0 stars, 0 forks, and 0 watchers. It has 1 branch (main) and 0 tags. The 'main' branch is selected, and the 'Code' button is highlighted. Below the repository information, there is a commit by 'kiru1711' titled 'Initial commit' with the hash 'bc45ae7' and a timestamp of '8 minutes ago'. The commit includes a file named 'README.md'. The 'README' file is expanded, showing the title 'Demo Project'.

```
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ explorer .

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git add .
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git commit -m"Updated readme"
[main fddedd2] Updated readme
 1 file changed, 4 insertions(+)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 312 bytes | 312.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kiru1711/DemoProject.git
   bc45ae7..fddedd2  main -> main
```

HANDS ON 2

```
MINGW64:/c:/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ echo "Git practice log" > sample.log

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ touch .gitignore

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ notepad .gitignore

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .gitignore

nothing added to commit but untracked files present (use "git add" to track)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git add .gitignore

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git commit -m "Added .gitignore file"
[main acd3eea] Added .gitignore file
1 file changed, 1 insertion(+)
create mode 100644 .gitignore

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 289 bytes | 289.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kiru1711/DemoProject.git
 fddedd2..acd3eea  main -> main

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git branch
```

```

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git branch
* main

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git branch GitNewBranch

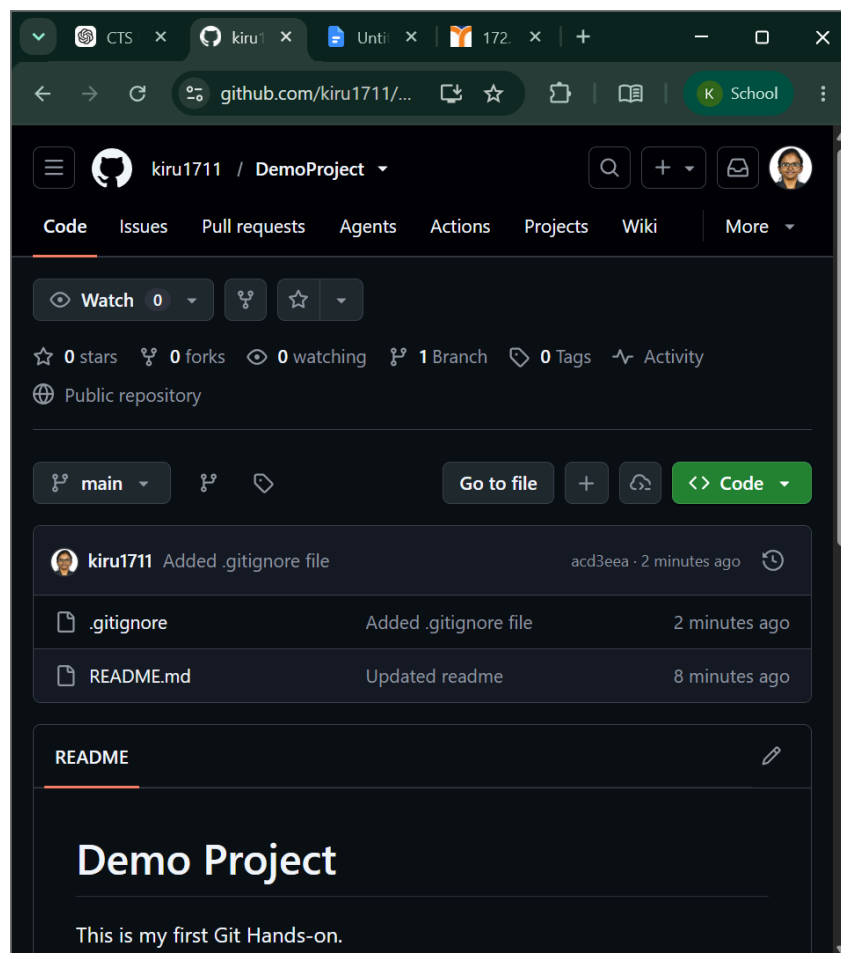
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git branch
  GitNewBranch
* main

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (GitNewBranch)
$ git branch
* GitNewBranch
  main

```

OUTPUT :



HANDS ON 3

```
MINGW64:/c/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git branch Gitwork

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (main)
$ git checkout Gitwork
Switched to branch 'Gitwork'

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ git branch
* Gitwork
  main

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ echo "<message>Hello from Gitwork branch</message>" > hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ cat hello.xml
<message>Hello from Gitwork branch</message>

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next ti
me Git touches it

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ git commit -m "Added hello.xml"
[Gitwork 07b3e59] Added hello.xml
1 file changed, 1 insertion(+)
create mode 100644 hello.xml
```

```
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (Gitwork)
$ git push -u origin Gitwork
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 312 bytes | 312.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'Gitwork' on GitHub by visiting:
remote:   https://github.com/kiru1711/DemoProject/pull/new/Gitwork
remote:
To https://github.com/kiru1711/DemoProject.git
 * [new branch]      Gitwork -> Gitwork
branch 'Gitwork' set up to track 'origin/Gitwork'.
```

OUTPUT :

The screenshot shows the GitHub web interface for a repository named 'DemoProject' by user 'kiru1711'. The 'Branches' tab is active, showing a list of branches. The 'main' branch is the default, updated 5 minutes ago. The 'Gitwork' branch is also listed, updated 1 minute ago, with 0 commits behind and 1 commit ahead. A 'New branch' button is visible in the top right corner of the branches section.

Branches [New branch](#)

[Overview](#) [Yours](#) [Active](#) [Stale](#) [All](#)

Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull
main	5 minutes ago				Default

Your branches

Branch	Updated	Check status	Behind	Ahead
Gitwork	1 minute ago		0	1

Active branches

Branch	Updated	Check status	Behind	Ahead
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HANDS ON 4

```
MINGW64; c:/Users/Kiruthika.R.J/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (Gitwork)
$ notepad hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (Gitwork)
$ git add hello.xml
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (Gitwork)
$ git commit -m"modified hello.xml"
[Gitwork 4037273] modified hello.xml
1 file changed, 1 insertion(+), 1 deletion(-)

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (Gitwork)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ notepad hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ notepad hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ echo "<message>Hello from Main branch</message>" > hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git add .
warning: in the working copy of 'hello.xml', LF will be replaced by CRLF the next time Git touches it

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git commit -m"Modified hello.xml in main branch"
[main 482345b] Modified hello.xml in main branch
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/DemoProject (main)
$ git log --oneline --graph --decorate --all
```

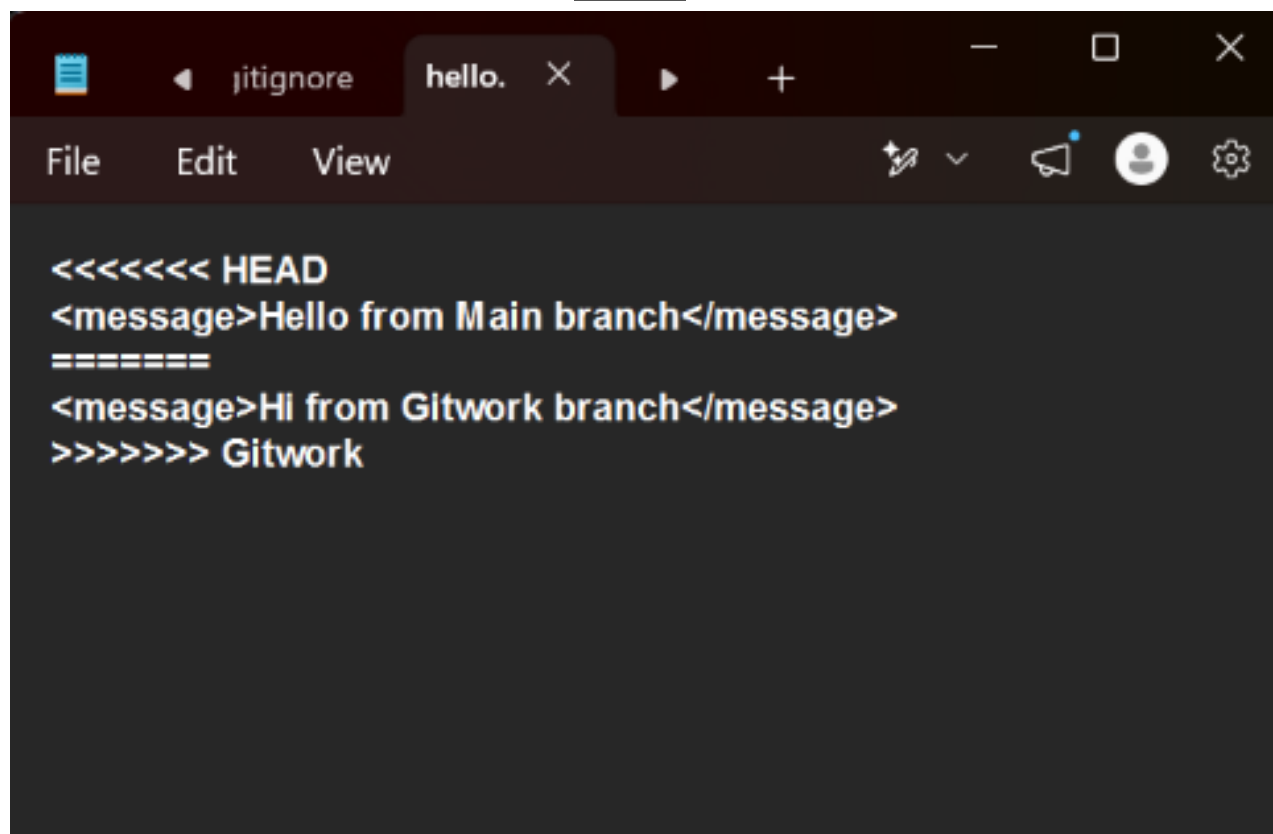
```

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (main)
$ git log --oneline --graph --decorate --all
* 482345b (HEAD -> main) Modified hello.xml in main branch
| * 4037273 (Gitwork) modified hello.xml
| * 07b3e59 (origin/Gitwork) Added hello.xml
|/
* 74f9aac (origin/main, origin/HEAD) Added branchFile in NewBranch
* acd3eea Added .gitignore file
* fddedd2 Updated readme
* bc45ae7 Initial commit

Kiruthika.R.J@LAPTOP-ULJRV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/De
moProject (main)
$ git merge Gitwork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

```

Conflict



Resolved

```
students.csv  c++  guidev  +
File Edit View

<message>Hello from Main and Gitwork branches</message>

MINGW64/c/Users/Kiruthika.RJ/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
* 74f9aac (origin/main, origin/HEAD) Added branchFile in NewBranch
* acd3eaa Added .gitignore file
* fddedd2 updated readme
* bc45ae7 Initial commit

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git merge Gitwork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main|MERGING)
$ <message>Hello from Main and Gitwork branches</message>
bash: syntax error near unexpected token `newline'

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main|MERGING)
$ git add hello.xml

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main|MERGING)
$ git commit -m "Resolved Merge Conflict"
[main f17a531] Resolved Merge Conflict

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 875 bytes | 437.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/Kirui1711/DemoProject.git
74f9aac..f17a531 main -> main

Kiruthika.R.3@LAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$
```

HANDS ON 5

The image shows a terminal window on the left and a GitHub repository page on the right, illustrating the process of creating and pushing a repository.

Terminal Window (MINGW64):

```
kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
$ AC

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ echo "Hello GitHub" > test.txt

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ ls
README.md branchFile.txt hello.xml sample.log test.txt

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git add test.txt
warning: in the working copy of 'test.txt', LF will be replaced by CRLF the next time
e Git touches it

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git commit -m "Added test.txt"
[main becc943] Added test.txt
1 file changed, 1 insertion(+)
create mode 100644 test.txt

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 286 bytes | 286.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/kiru1711/DemoProject.git
 f17a531..becc943  main -> main

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$

kiruthika.R.3BLAPTOP-UL3RV6PK MINGW64 ~/IdeaProjects/LibraryManagement/GitHandsOn/Demo...
moProject (main)
$ |
```

GitHub Repository Page (github.com/kiru1711/DemoProject):

The page shows the repository details for **kiru1711 / DemoProject**. It includes a table of recent commits:

Commit	Message	Time
becc943	Added test.txt	now
f17a531	Added .gitignore file	39 minutes ago
becc943	Updated readme	45 minutes ago
f17a531	Added branchFile in NewBranch	32 minutes ago
f17a531	Resolved Merge Conflict	15 minutes ago
f17a531	Added test.txt	now

The repository also shows a list of files: **.gitignore**, **README.md**, **branchFile.txt**, **hello.xml**, and **test.txt**.